

VIETNAM NATIONAL UNIVERSITY – HCMC UNIVERSITY OF SCIENCE

Faculty of Information and Technology

Software Engineering Department



Testing

Project name: Master Interview

Course: Introduction to Software Engineering

Class: 24C10

Students:

24127186 - Giáp Đăng Khoa

24127518 - Trần Ngọc Quân

24127136 - Nguyễn Thanh Trọng

24127137 - Nguyễn Thanh Trúc

Lecturers:

Nguyễn Minh Huy

Ngô Ngọc Đăng Khoa

Trương Phước Lộc

Table of Contents

Objectives	1
1 Member Contribution Assessment	2
2 Test plan	6
3 Test cases	8
<i>1.1 List of test cases</i>	<i>8</i>
<i>1.2 Test case specifications</i>	<i>9</i>
4 AI Usage Declaration	32
5 Presentation	34
6 Reflective Report	35

Software Testing

Objectives

This document focus on the following topics:

- ✓ Completing the Software Testing document with the following sections:
 - Test Plan
 - Test Cases
- ✓ Understanding the Software Testing document.
- ✓ This document will be used as input for AI tools to verify the quality of subsequent project artifacts.

All project artifacts must remain consistent and synchronized.

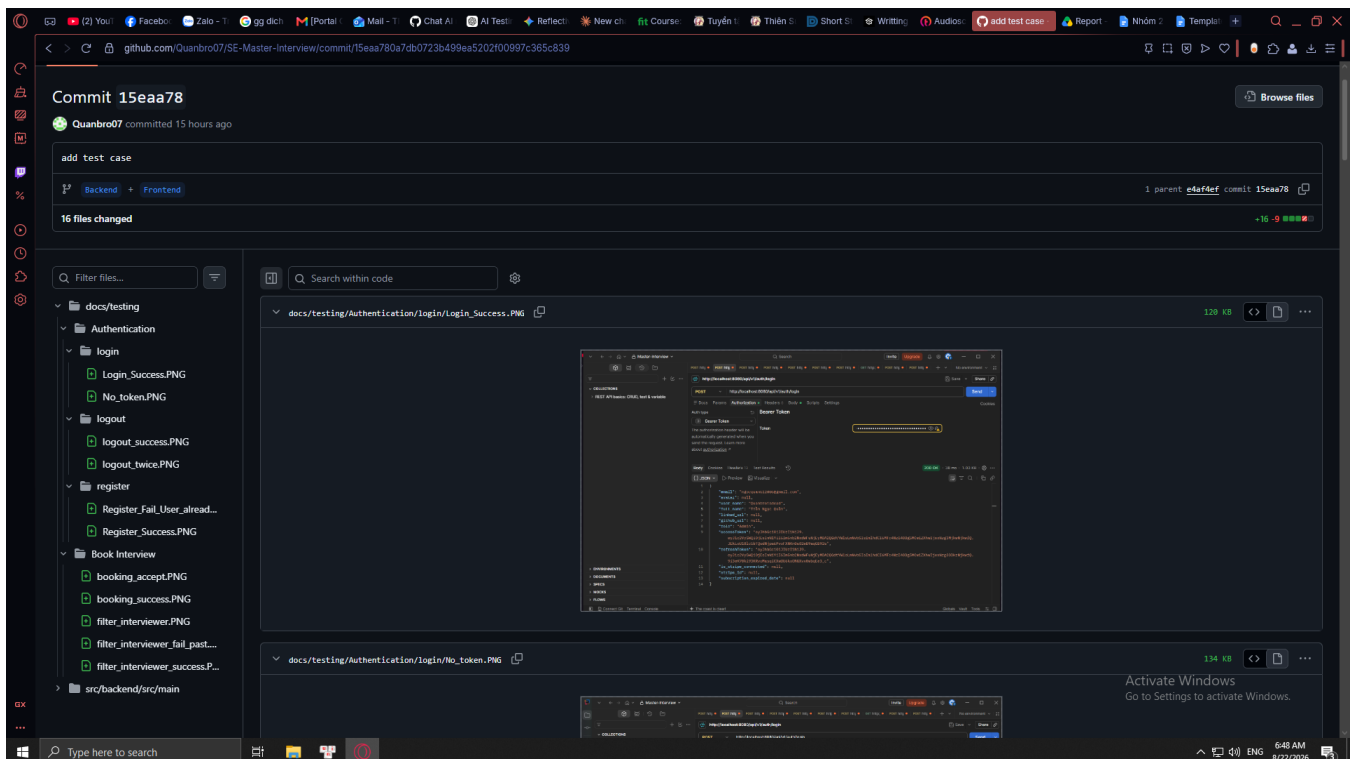
For example, if the project proposal is modified during this phase, a new version of the proposal must be documented.

By the conclusion of the project, all versions of every artifact must be submitted to demonstrate the evolution of your work.

1 Member Contribution Assessment

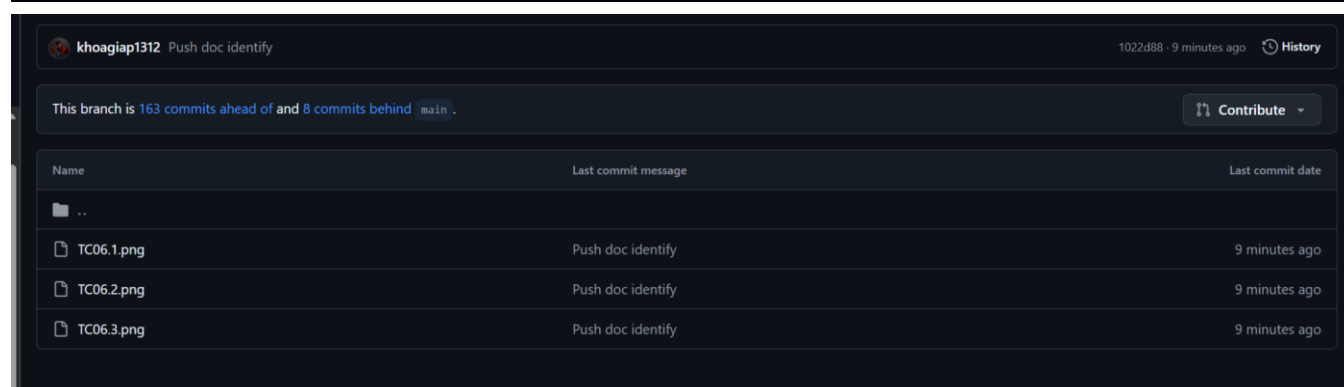
24127518 - Trần Ngọc Quân - Contribution (25%)

Task name	Summary
Testing: Authentication(Login, Register, Logout) and Booking	Perform manual testing and validate edge cases for user authentication modules (Login, Register, Logout).
Write report: Test Case Authentication and Booking	Document test scenarios, execution results, and defects for the Authentication and Booking module



24127186 - Giáp Đăng Khoa- Contribution (25%)

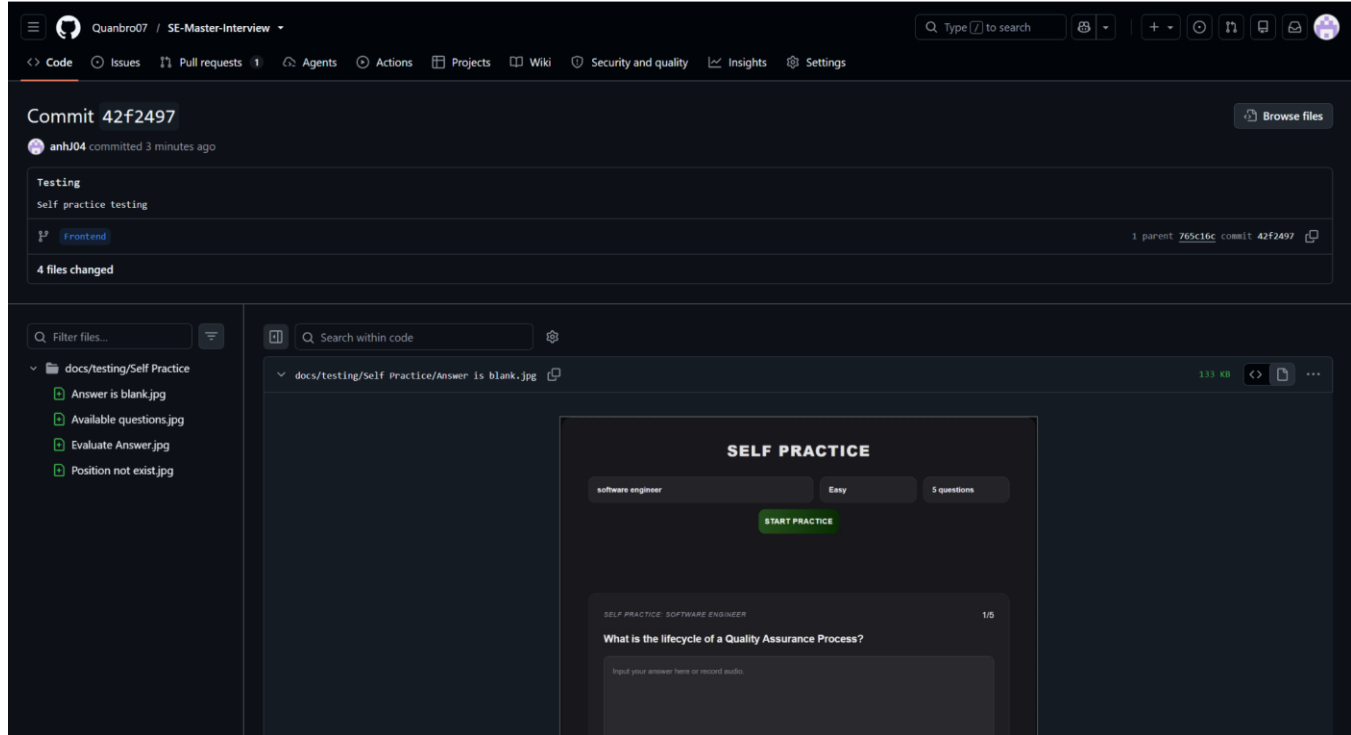
Task name	Summary
Testing: CV Assessment, MCP Server	Executed manual test cases for both modules (Normal & Edge cases). Verified file upload logic (CV Assessment) and API response accuracy (MCP Server). Logged actual results and identified potential bugs/edge cases.
Write Report: Test Case for CV Assessment, MCP Server	Drafted detailed test case specifications in English. Defined clear contexts, input data, test steps, and expected outputs for all scenarios within the CV Assessment and MCP Server features.



24127136 - Nguyễn Thanh Trọng - Contribution (25%)

Task name	Summary
Testing: Self practice	Conduct functional testing for Self Practice feature including fetching questions, position validation, AI answer evaluation, and blank input handling.
Write Report: Test Case for Self Practice	Document test case report covering TC-03.1 (Interviewer's Feedback), TC-03.2 (User Absence), TC-05.1 (Fetch Questions), TC-05.2

(Non-existent Position), TC-05.3 (Answer Evaluation), and TC-05.4 (Blank Answer Handling).



24127137- Nguyễn Thanh Trúc - Contribution (25%)

Task name	Summary
Design: Test case Plan	Design a full test plan with various test techniques.
Write report: Test case plan	Documenting a test cases list covers most critical problems.

The screenshot shows a GitHub commit page for the repository 'SE-Master-Interview'. The commit is titled 'Commit 565700d' and was made by 'Nguyễn Thanh Trúc' 1 minute ago. The commit message is 'test plan and test case list'. The commit is on the 'main' branch and has one parent commit 'ae42844'. A file named 'testplan.pdf' is shown as changed, with a size of 216 KB. The file is located in the 'docs/testing' directory. The page includes a search bar, a file browser, and a comments section.

2

Test plan

Please include the ID and full name of the student who authored, reviewed, or updated this section.

Written by: 24127137 - Nguyễn Thanh Trúc

Edited by:

Reviewed by:

Seq	Test type	Objects	Techniques
1	Static testing	System requirement specification	White box: Review if the product features match the requirements
		Design documentation	
2	Unit testing	Authentication	Black box: OAuth2 on Google, Github Whitebox: Verify JWT generation
		Self-practice question function	White box: Retrieve questions from the database based on positions, difficulty, and number of questions.
		Answer evaluation function	White box: Evaluation user answer based on the matching keyword
		Booking interview function	White box: Booked available schedule on the website
		MCP server	Functional testing: search questions, view schedule. Non-functional testing: Performance test: Agent reply time.
		External service functions	White-box: Access external service API.

3	Integration testing	Database, cache integration	Gray-box: The backend can recall data from the database and cache.
		External service navigation	Gray-box: Under pre-condition, service returns API response
		Backend API	Black-box: Verify backend API gateways
4	System testing	Complete user flow	Black box: Verify that the images build, environment variables map correctly, and containers communicate on the network.

3

Test cases

1.1 List of test cases

Please include the ID and full name of the student who authored, reviewed, or updated this section.

Written by: 24127137 - Nguyễn Thanh TrúC

Edited by:

Reviewed by: 24127518 - Trần Ngọc Quân, 24127186 - Giáp Đăng Khoa, 24127136 - Nguyễn Thanh Trọng, 241271367 - Nguyễn Thanh TrúC

Seq	Test case	Feature	Description
TC-01.1	Register: Github, Google	Authentication	Generate access token, refresh token
TC-01.2	Log in		Validate access token
TC-01.3	Log out		Black listed token
TC-02.1	Normal case: Interviewee book interview	Book interview	Redirect to payment page
TC-02.2	Interviewee doesn't pay		Payment timeout
TC-02.3	Interviewer accept request		Request payment and generate Zoom link
TC-02.4	Interviewer decline request		Update schedule status.
TC-03.1	Normal case: Interview - feedback	Mock interview	Send feedback to interviewee.
TC-03.2	Edge case: Interviewer/Interviewee absence		Zoom links expire at the scheduled end time.

TC-04	Normal case: User send CV	CV assessment	Evaluate CV based on job position
TC-04	Edge case: User send blank CV		CV receive 0 score
TC-05.1	Self-practice questions	Self practice	Retrieve questions from the database based on positions, difficulty, and number of questions.
TC-05.2	Positions not exist		Response none question
TC-05.3	Evaluate self-practice answers		Evaluation user answer based on the matching keyword
TC-05.4	Answer is blank		Do not evaluate when the answer is blank.
TC-06.1	Get practice questions by positions	MCP server	Retrieve questions from the database based on positions, difficulty, and number of questions by MCP tools
TC-06.2	Get available schedule by position and date		Receive list of available schedule
TC-06.3	Out of scope question		No reply

1.2 Test case specifications

Written by: 24127518 - Trần Ngọc Quân, 24127186 - Giáp Đăng Khoa, 24127136 - Nguyễn Thanh Trọng, 241271367 - Nguyễn Thanh Trúc

Edited by:

Reviewed by: 24127518 - Trần Ngọc Quân, 24127186 - Giáp Đăng Khoa, 24127136 - Nguyễn Thanh Trọng, 241271367 - Nguyễn Thanh Trúc.

1.2.1 TC-01: Authentication

[illegible]

Logout:

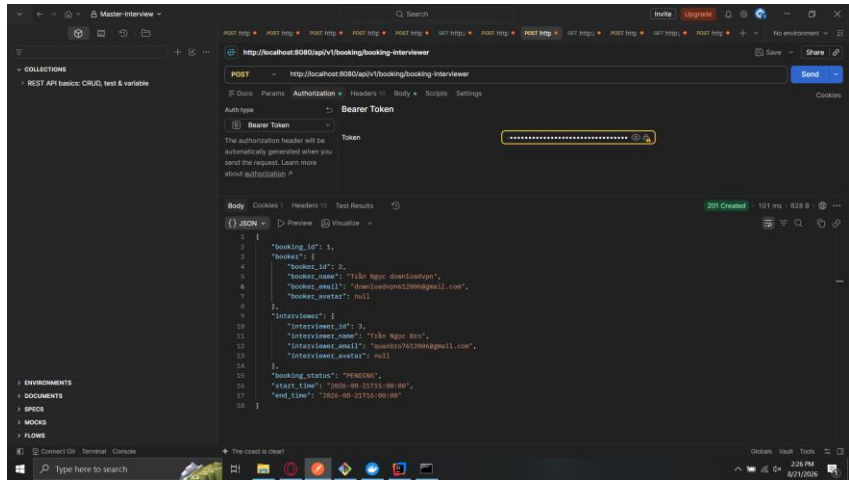
The first screenshot shows a POST request to `http://localhost:8080/api/v1/auth/register` with a Bearer Token authorization header. The response is a 200 OK status with a JSON body containing user details like email, avatar, and role.

The second screenshot shows a POST request to `http://localhost:8080/api/v1/auth/logout` with a Bearer Token authorization header. The response is a 200 OK status with a body of `{ "done": true }`.

The third screenshot shows a POST request to `http://localhost:8080/api/v1/auth/logout` with a Bearer Token authorization header. The response is a 401 Unauthorized status with a JSON body indicating the token has expired and login is required.

Result	Passed
--------	--------

1.2.2 TC-02.1. Interviewee book interviewer

Test case	TC-02.1
Related feature	Booking Interviewer
Context	After interviewee view interviewer information. The interviewee decided to book the interviewer.
Input Data	interviewer id, position name, start date, end date, note
Expected Output	Booking successfully, return Response for frontend to display
Test steps	Choose interviewer id to book
Actual Output	 <pre> 1 { 2 "booking_id": 1, 3 "booker": { 4 "booker_id": 2, 5 "booker_name": "Tisha Ngoc Thanh", 6 "booker_email": "tishanngoc120@gmail.com", 7 "booker_status": null 8 }, 9 "interviewer": { 10 "interviewer_id": 3, 11 "interviewer_name": "Tisha Ngoc Bui", 12 "interviewer_email": "tishanngoc120@gmail.com", 13 "interviewer_status": null 14 }, 15 "booking_status": "PENDING", 16 "start_time": "2024-09-22T13:00:00", 17 "end_time": "2024-09-22T13:00:00" 18 } </pre>
Result	Passed

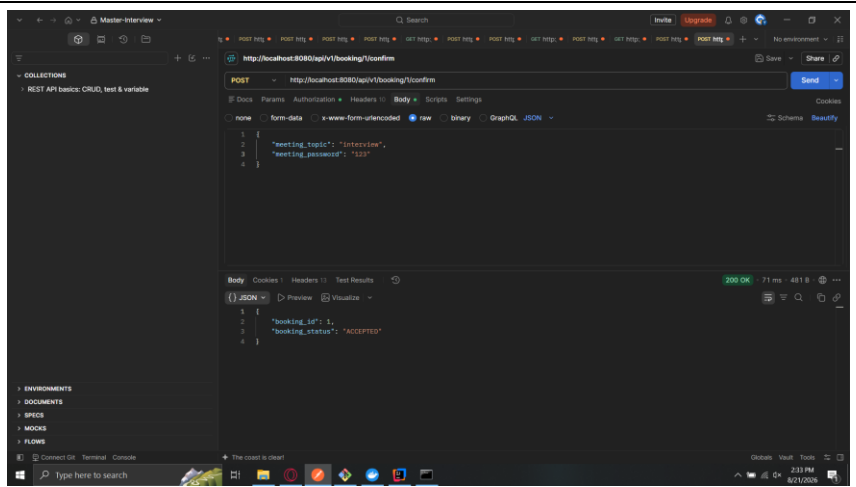
1.2.3 TC-02.2 Interviewee does not pay

Test case	TC-02.2
Related feature	Interviewee doesn't pay
Context	After the interviewer confirms the booking interview but interviewee never pays to process the next stage.
Input Data	Booking status at 'CONFIRM'

Expected Output	Release the blocked time slot for the interviewer and remove the interviewee from the booking.
Test steps	User books interviewer Interviewer accepts interview booking User does not pay
Actual Output	The interviewee still held the time slot and the interviewer lost that time slot.
Result	Failed

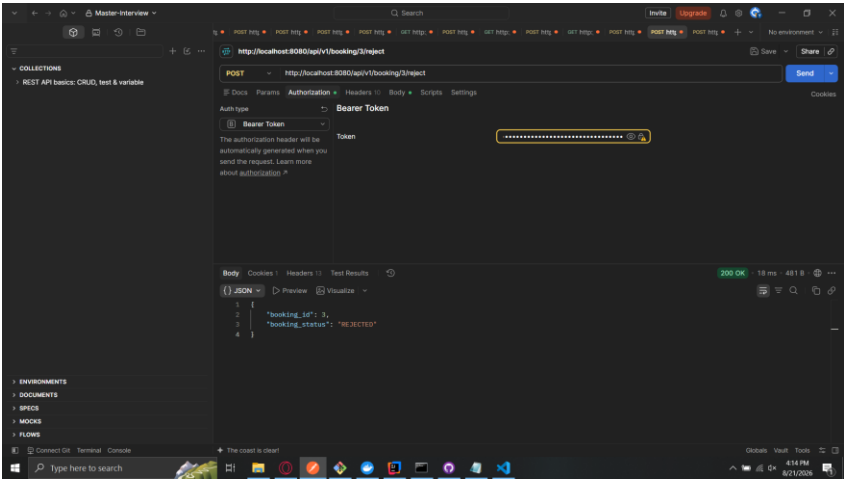
1.2.4 TC-02.3 Interviewer accept request

<i>Test case</i>	TC-02.3
Related feature	Interviewer accept request
Context	After interviewee booking interviewer. The Interviewer accepts the interview section.
Input Data	Booking id, meeting topic, meeting password
Expected Output	Return response with booking id and booking status 'Accepted'
Test steps	User books interviewer Interviewer accepts interview booking

Actual Output	
Result	Passed

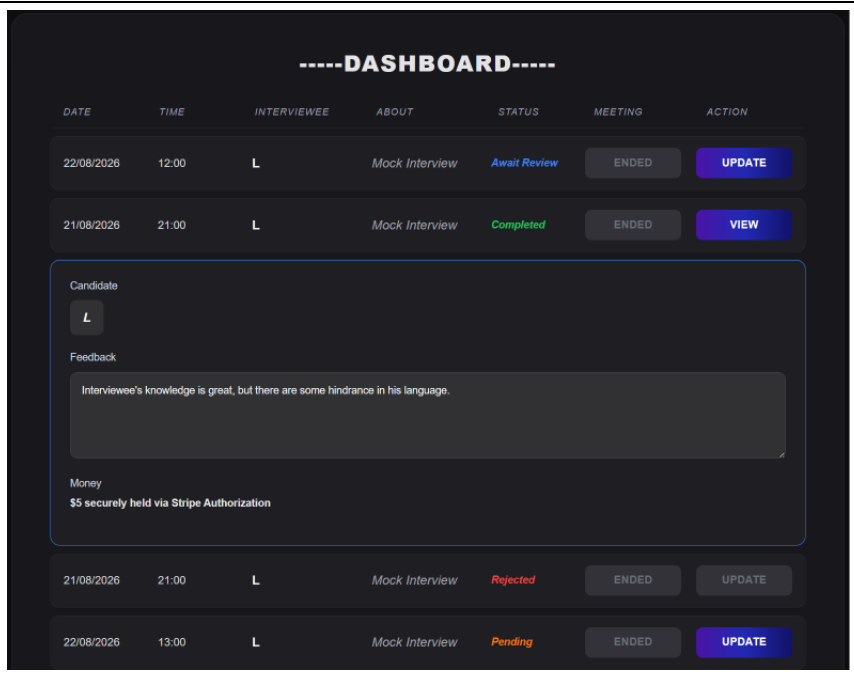
1.2.5 TC-02.4 Interviewer decline request

Test case	TC-02.4
Related feature	Interviewer decline request
Context	After interviewee booking interviewer. The Interviewer accepts the interview section.
Input Data	Booking id, meeting topic, meeting password
Expected Output	Return response with booking id and booking status 'Decline'
Test steps	User books interviewer Interviewer decline request

Actual Output	
Result	Passed

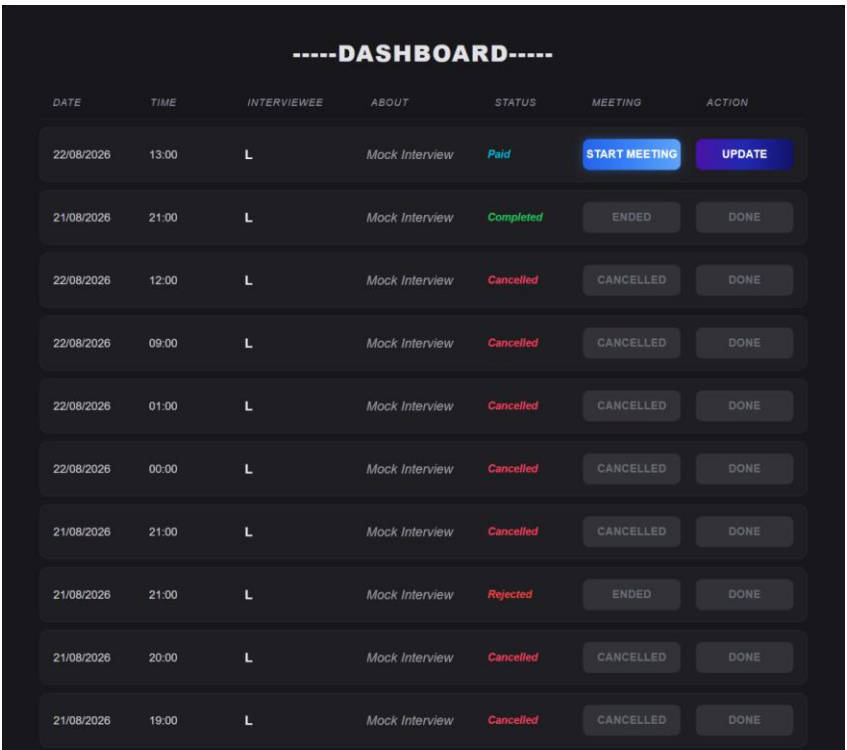
1.2.6 TC-03.1 Interviewer's feedback

Test case	TC-03.1
Related feature	Interviewer's feedback
Context	After a booking is complete, an interviewer can send feedback
Input Data	Meeting complete Feedback input
Expected Output	Booking status changes to 'Complete'
Test steps	Enter Zoom meeting End meeting Write feedback Press Complete

Actual Output	
Result	Passed

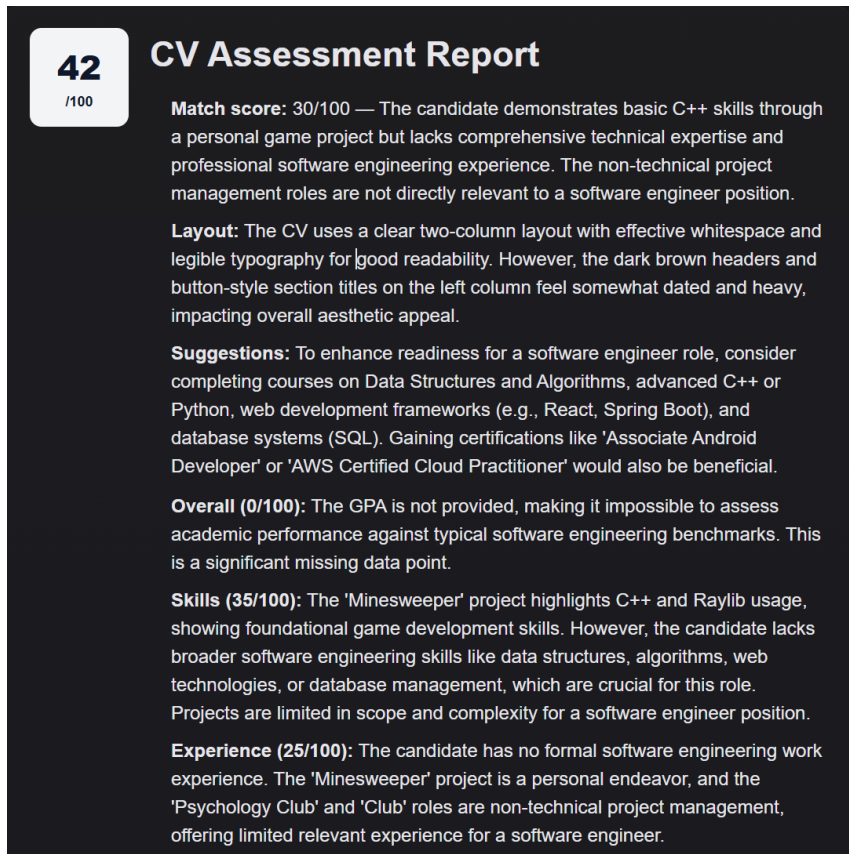
1.2.7 TC-03.2 Interviewee/Interviewer absence

Test case	TC-03.2
Related feature	Interviewee/Interviewer absence
Context	Interviewee or (and) interviewer doesn't join the meeting, zoom links expired, meeting buttons disabled
Input Data	None
Expected Output	Disable buttons
Test steps	Book a meeting Not joining after the end-time has passed

Actual Output	
Result	Passed

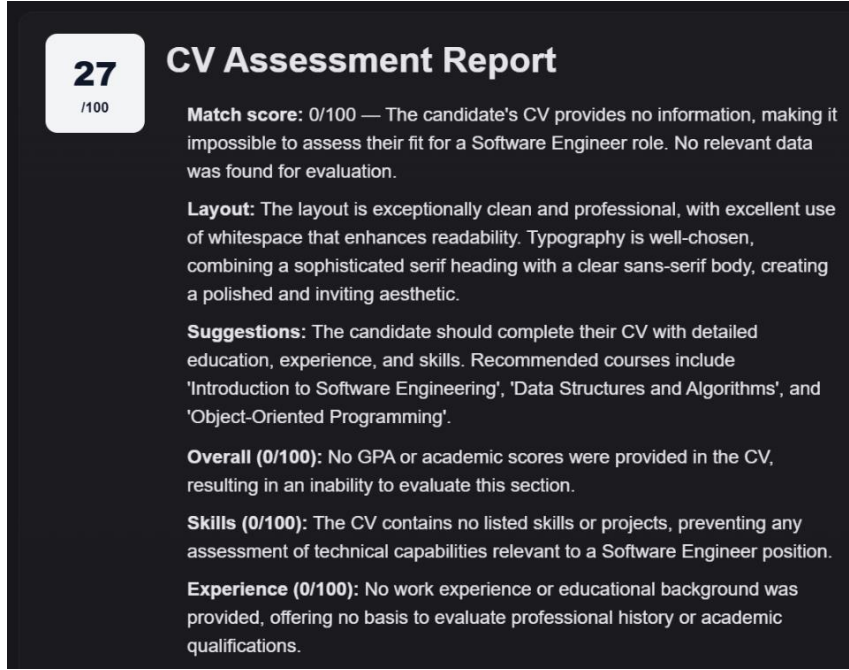
1.2.8 TC04.1(Normal case: user send a CV)

Test case	TC-04.1
Related feature	CV Assessment
Context	The user (Interviewee) is logged into the system and is currently on the CV upload page. The system has active job positions available for selection.
Input Data	A valid CV file containing readable text (e.g., standard .PDF or .DOCX format). A selected Job Position. In this case, input is a CV relevant to a Software Engineer.
Expected Output	The evaluation score and relevant feedback.
Test steps	Select a specific Job Position from the dropdown or list. Click the "Upload CV" button.

	Choose a valid CV file from the local directory and complete the upload. Click the "Submit" (or "Evaluate") button.
Actual Output	 42 /100 CV Assessment Report Match score: 30/100 — The candidate demonstrates basic C++ skills through a personal game project but lacks comprehensive technical expertise and professional software engineering experience. The non-technical project management roles are not directly relevant to a software engineer position. Layout: The CV uses a clear two-column layout with effective whitespace and legible typography for good readability. However, the dark brown headers and button-style section titles on the left column feel somewhat dated and heavy, impacting overall aesthetic appeal. Suggestions: To enhance readiness for a software engineer role, consider completing courses on Data Structures and Algorithms, advanced C++ or Python, web development frameworks (e.g., React, Spring Boot), and database systems (SQL). Gaining certifications like 'Associate Android Developer' or 'AWS Certified Cloud Practitioner' would also be beneficial. Overall (0/100): The GPA is not provided, making it impossible to assess academic performance against typical software engineering benchmarks. This is a significant missing data point. Skills (35/100): The 'Minesweeper' project highlights C++ and Raylib usage, showing foundational game development skills. However, the candidate lacks broader software engineering skills like data structures, algorithms, web technologies, or database management, which are crucial for this role. Projects are limited in scope and complexity for a software engineer position. Experience (25/100): The candidate has no formal software engineering work experience. The 'Minesweeper' project is a personal endeavor, and the 'Psychology Club' and 'Club' roles are non-technical project management, offering limited relevant experience for a software engineer.
Result	Passed

1.2.9 TC-04.2(User send a blank CV)

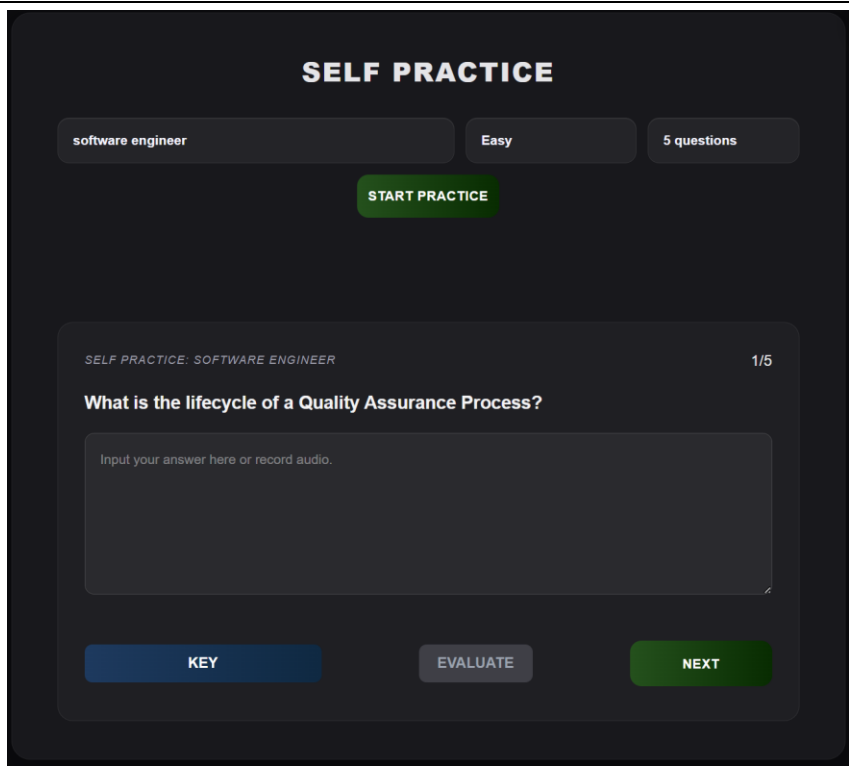
Test case	TC-04.2
Related feature	CV Assessment
Context	The user (Interviewee) is logged into the system and is currently on the CV upload page. The system has active job positions available for selection.
Input Data	A valid CV file containing readable text (e.g., standard .PDF or .DOCX format). A selected Job Position. In this case, it is a blank CV or irrelevant to a Software Engineer.

Expected Output	The system successfully receives the uploaded file. The system evaluates the CV based on the selected job position requirements. The system successfully returns a specific evaluation score and relevant feedback.
Test steps	Select a specific Job Position from the dropdown or list. Click the "Upload CV" button. Choose a valid CV file from the local directory and complete the upload. Click the "Submit" (or "Evaluate") button.
Actual Output	 27 /100 CV Assessment Report Match score: 0/100 — The candidate's CV provides no information, making it impossible to assess their fit for a Software Engineer role. No relevant data was found for evaluation. Layout: The layout is exceptionally clean and professional, with excellent use of whitespace that enhances readability. Typography is well-chosen, combining a sophisticated serif heading with a clear sans-serif body, creating a polished and inviting aesthetic. Suggestions: The candidate should complete their CV with detailed education, experience, and skills. Recommended courses include 'Introduction to Software Engineering', 'Data Structures and Algorithms', and 'Object-Oriented Programming'. Overall (0/100): No GPA or academic scores were provided in the CV, resulting in an inability to evaluate this section. Skills (0/100): The CV contains no listed skills or projects, preventing any assessment of technical capabilities relevant to a Software Engineer position. Experience (0/100): No work experience or educational background was provided, offering no basis to evaluate professional history or academic qualifications.
Result	Passed

1.2.10 TC-05.1 (Selfpractice's questions)

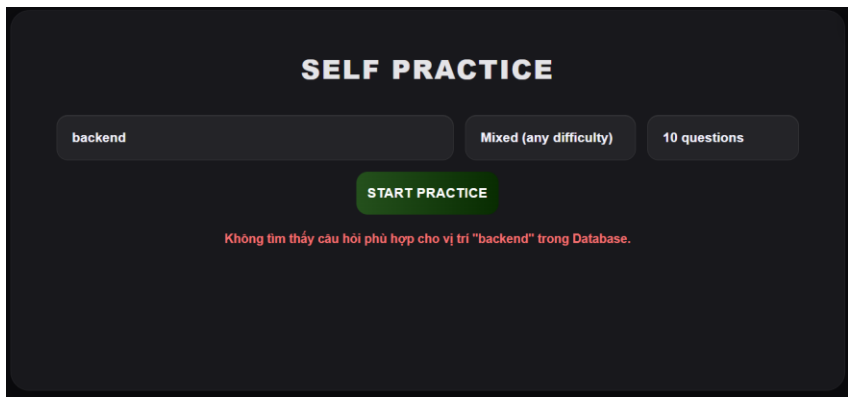
Test case	TC-05.1
Related feature	Self Practice

Context	The user (Interviewee) is logged into the system and is currently on the Self Practice page. The system allows selecting position, difficulty level, and total number of questions to practice.
Input Data	Position: "software engineer" Difficulty Level: Easy Question Count: 5 questions
Expected Output	The system successfully queries the database and fetches the requested list of practice questions based on the selected criteria (Position: software engineer, Difficulty: Easy, Total: 5). The system renders the first question ("What is the lifecycle of a Quality Assurance Process?") along with the question index (1/5). An answer input field (text/audio) and interactive buttons ("KEY", "EVALUATE", "NEXT") are displayed properly for the user to start practicing.
Test steps	Select or enter "software engineer" as the job position. Select "Easy" as the difficulty level. Select "5 questions" as the total question count. Click the "START PRACTICE" button to send the request to fetch questions.

Actual Output	
Result	Passed

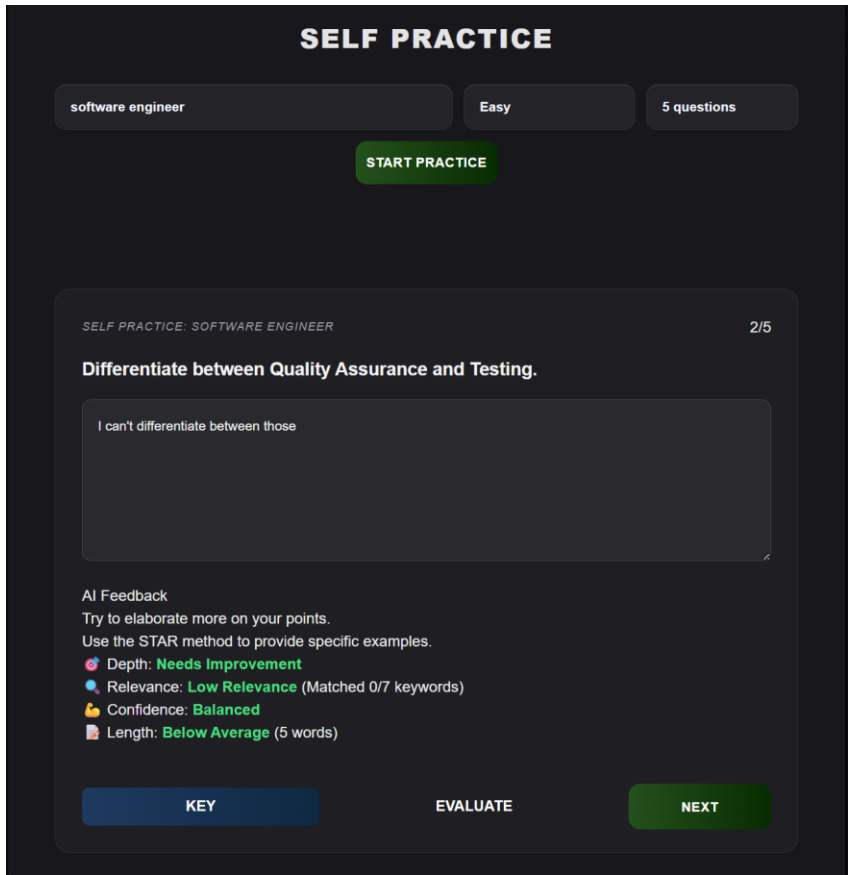
1.2.11 TC-05.2(Position not exist)

Test case	TC-05.2
Related feature	Self Practice
Context	The user (Interviewee) is logged into the system and is currently on the Self Practice configuration page. The user attempts to start a practice session by entering or selecting a job position that does not exist in the database.
Input Data	Position: "backend" (a position that has no questions stored in the database) Difficulty Level: Mixed (any difficulty) Question Count: 10 questions
Expected Output	The system prevents the practice session from starting.

	The system displays an error message below the form stating: "Không tìm thấy câu hỏi phù hợp cho vị trí "backend" trong Database." No practice question panel is rendered.
Test steps	Enter or select an invalid/non-existent position name (e.g., "backend") in the position input field. Select difficulty level (e.g., "Mixed (any difficulty)"). Select total questions (e.g., "10 questions"). Click the "START PRACTICE" button.
Actual Output	 The screenshot shows a dark-themed interface titled "SELF PRACTICE". It features three input fields: "backend", "Mixed (any difficulty)", and "10 questions". Below these fields is a green "START PRACTICE" button. A red error message is displayed at the bottom: "Không tìm thấy câu hỏi phù hợp cho vị trí "backend" trong Database."
Result	Passed

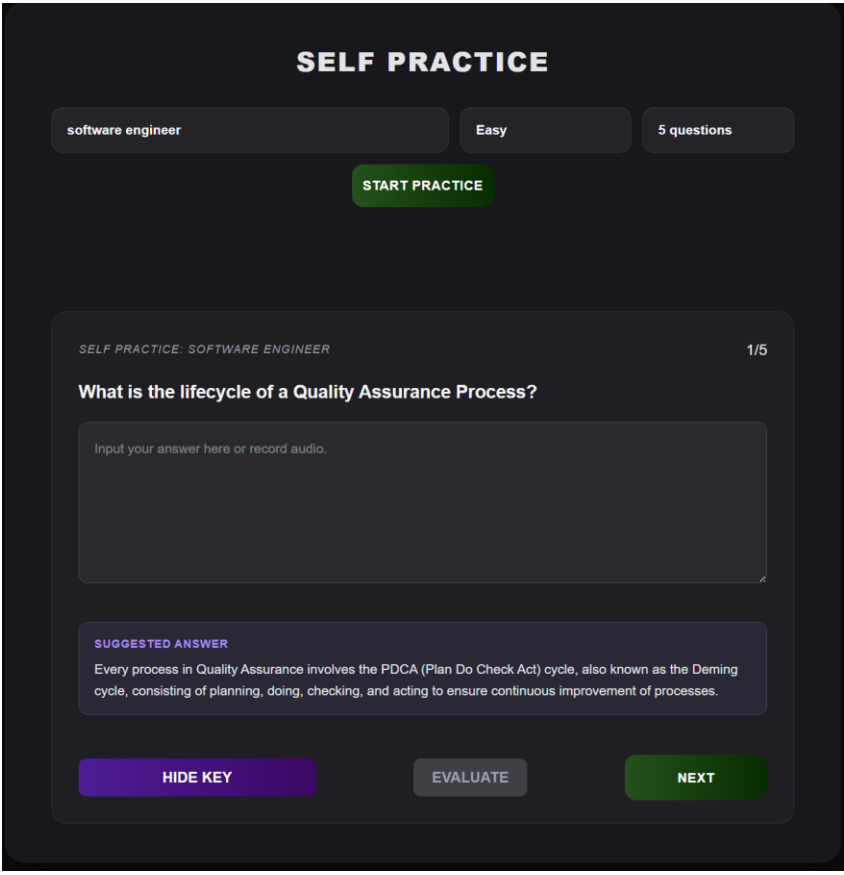
1.2.12 TC-05.3(Evaluate answer)

Test case	TC-05.3
Related feature	Self Practice
Context	The user (Interviewee) is in an active Self Practice session and has typed an answer to a practice question. The user triggers the evaluation feature to receive detailed feedback from the AI.
Input Data	Position: Software Engineer Difficulty Level: Easy Question: "Differentiate between Quality Assurance and Testing." (Question 2/5) Answer Input: "I can't differentiate between those"
Expected Output	The system processes the submitted answer and displays the "AI Feedback" section directly below the input area.

	The feedback section includes suggestions for improvement (e.g., "Try to elaborate more on your points", "Use the STAR method to provide specific examples."). The system returns detailed evaluation metrics: ● Depth: Needs Improvement ● Relevance: Low Relevance (Matched 0/7 keywords) ● Confidence: Balanced ● Length: Below Average (5 words)
Test steps	Navigate to question 2 of the "Software Engineer - Easy" practice session. Enter the response "I can't differentiate between those" in the text area. Click the "EVALUATE" button.
Actual Output	 The screenshot displays the 'SELF PRACTICE' interface. At the top, it shows 'software engineer', 'Easy', and '5 questions' with a 'START PRACTICE' button. Below, the question is 'Differentiate between Quality Assurance and Testing.' The user's response is 'I can't differentiate between those'. The AI Feedback section provides the following metrics: Depth: Needs Improvement, Relevance: Low Relevance (Matched 0/7 keywords), Confidence: Balanced, and Length: Below Average (5 words). At the bottom, there are buttons for 'KEY', 'EVALUATE', and 'NEXT'.
Result	Passed

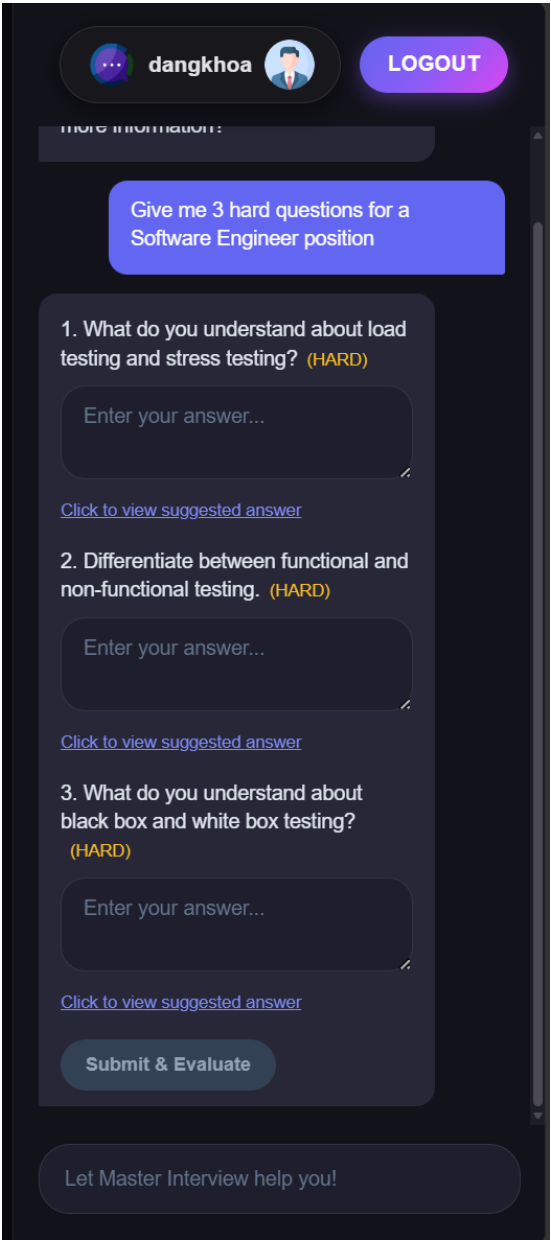
1.2.13 TC-05.4(Answer is blank)

Test case	TC-05.4
Related feature	Self Practice
Context	The user (Interviewee) is in an active practice session and leaves the answer input field completely empty. The user wants to view the suggested answer/key or attempt to evaluate without entering any input.
Input Data	Position: "software engineer" Difficulty Level: Easy Question: "What is the lifecycle of a Quality Assurance Process?" (Question 1/5) Answer Input: Blank / Empty
Expected Output	The "EVALUATE" button remains disabled (greyed out and unclickable) because no answer has been provided. Upon clicking the "KEY" button: • A "SUGGESTED ANSWER" panel appears below the text box displaying the reference answer (e.g., PDCA cycle). • The button label toggles from "KEY" to "HIDE KEY" (highlighted in purple).
Test steps	Start a practice session for "software engineer" with "Easy" difficulty. Leave the answer text area ("Input your answer here or record audio.") blank. Verify that the "EVALUATE" button is disabled and cannot be clicked. Click the "KEY" button. Verify that the "SUGGESTED ANSWER" panel is displayed and the button changes to "HIDE KEY".

Actual Output	
Result	Passed

1.2.14 TC-06.1

<i>Test case</i>	TC-06.1(Get practice questions by positions)
Related feature	MCP Server
Context	The user is interacting with the system's chat interface (or API endpoint) that connects to the MCP server. The database contains a variety of questions categorized by position and difficulty.
Input Data	A prompt specifying three parameters: Job Position, Difficulty, and Number of Questions (e.g., "Give me 3 hard questions for a Backend Developer position").
Expected Output	The MCP tools successfully query the database based on the provided parameters. The system returns exactly the requested number of questions. The returned questions match the specified job position and difficulty level.
Test steps	Open the interface connected to the MCP server. Input the request specifying the position, difficulty, and quantity. Send the request. Verify the response returned by the server.

Actual Output	
Result	Passed

1.2.4. TC-06.2:

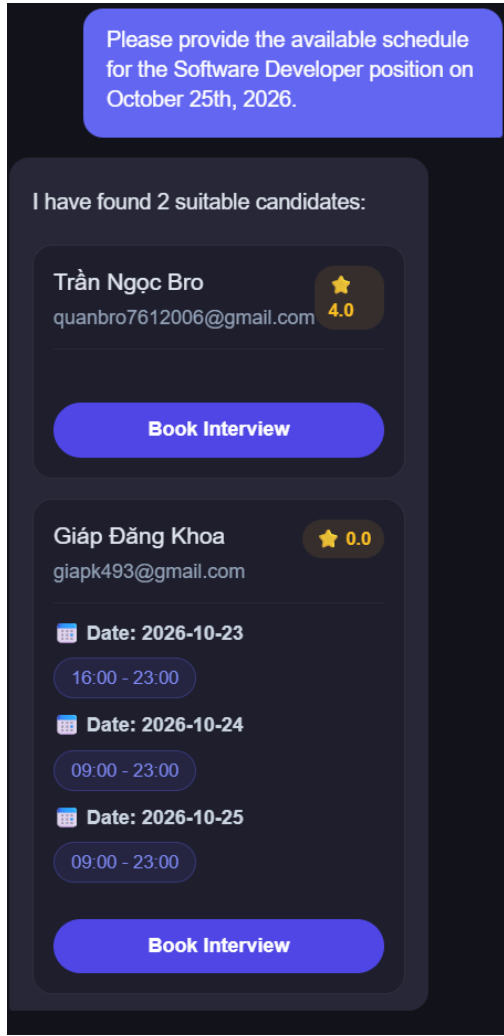
Please include the ID and full name of the student who authored, reviewed, or updated this section.

Written by: 24127186 - Giáp Đăng Khoa

Edited by:

Reviewed by:

<i>Test case</i>	TC-06.2(Get available schedule by position and date)
Related feature	MCP server
Context	The user is actively interacting with the MCP server chat interface. The system's scheduling database contains available interview slots for various roles.
Input Data	User prompt: <i>"Please provide the available schedule for the ReactJS Developer position on October 25th, 2026."</i>
Expected Output	The system successfully parses the parameters (Position: ReactJS Developer, Date: Oct 25, 2026) and returns an accurate list of available schedules.
Test steps	1. Access the MCP server interface. 2. Enter the defined prompt into the chat input. 3. Submit the request and wait for the response.

Actual Output	
Result	Passed

1.2.4. TC-06.3:

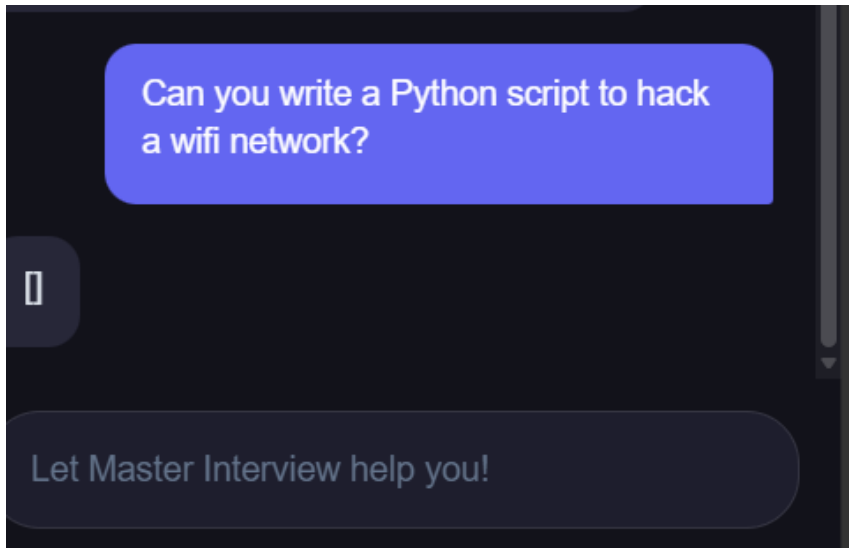
Please include the ID and full name of the student who authored, reviewed, or updated this section.

Written by: 24127186 - Giáp Đăng Khoa

Edited by:

Reviewed by:

Test case	TC-06.3(Out of scope question)
Related feature	MCP Server

Context	The user is actively interacting with the MCP server interface. The server is strictly configured to handle recruitment, scheduling, and technical practice domains.
Input Data	A prompt completely unrelated to the system's intended domain (e.g., "Can you write a Python script to hack a wifi network?" or "Give me a recipe for chocolate cake").
Expected Output	Null output.
Test steps	1. Access the MCP server chat or API interface. 2. Input the out-of-scope prompt. 3. Submit the request and observe the server's behavior.
Actual Output	 A screenshot of a chat interface with a dark background. A blue speech bubble contains the text "Can you write a Python script to hack a wifi network?". Below it, a dark grey speech bubble contains a placeholder icon (a square with a vertical line) and the text "Let Master Interview help you!".
Result	Passed

4

AI Usage Declaration

Based on the provided AI Usage Guideline document, teams must submit their AI usage declaration here. Students are encouraged to leverage AI tools creatively and effectively in this project.

Gemini. Gemini, Google, gemini.google.com, accessed [08:37 on August 21, 2026], prompt: “Based on the functional requirements and system architecture, provide suggestions for an appropriate testing plan and testing roadmap for the mock interview system”, used for the [test plan and testing strategy]; AI provided recommendations on testing phases, testing priorities, test levels, and the appropriate order for testing different system modules, students evaluated the suggestions and incorporated the suitable testing approaches into the project Test Plan.

Gemini. Gemini, Google, gemini.google.com, accessed [14:22 on August 21, 2026], prompt: “Based on the functional requirements of the mock interview system, identify possible edge cases and generate test cases that can verify these scenarios”, used for the [test case design]; AI suggested test cases covering normal conditions, boundary values, invalid inputs, booking conflicts, authentication failures, payment failures, and other exceptional scenarios, students reviewed and adapted the generated test cases to match the actual system implementation.

CGemini. Gemini, Google, gemini.google.com, accessed [15:14 on August 21, 2026], prompt: “Review the testing scope of this mock interview system and suggest additional areas that should be tested, including integration, security, reliability, and performance testing”, used for the [testing scope and coverage]; AI identified additional testing areas such as API integration, authorization, concurrent booking, transaction handling, email notification, system recovery, and response-time testing, students selected relevant testing scenarios and incorporated them into the overall testing strategy.

5

Presentation

Teams are required to record presentation videos of their work using the provided template.

Every team member must participate in the presentation, specifically covering the sections their contributed to the project.

Videos must not exceed 30 minutes.

Please upload your videos to Youtube (set to either Unlisted or Public).

Provide the Youtube links the the field below.

Here is the Youtube links:

<https://youtu.be/itVbP-T4hNA>

6

Reflective Report

Which sections of this template are most helpful for implementation, and which are unnecessary? Provide specific examples to support your reasoning.

Most Helpful for Implementation: Edge Case Scenarios (Specifically: *TC 02.2 Interviewee does not pay*)

- **Reasoning:** This section provides immense practical value by forcing the team to anticipate and handle unconventional user behaviors or negative paths that were not obvious during initial development. For instance, testing the edge case *TC 02.2 (Interviewee does not pay)* targets a critical flaw in the business logic. Anticipating this scenario ensures the development team implements proper system safeguards-such as preventing access to interview rooms or displaying specific error prompts-rather than just assuming the "happy path." This directly improves the application's overall robustness.

Unnecessary / Less Practical: The Formal Test Plan Document

- **Reasoning:** In the context of a small-scale academic project with a limited feature set, dedicating significant time to drafting a comprehensive, formal Test Plan lacks practical value. The heavy documentation overhead feels redundant because the scope is small enough that the team can transition directly into test execution (ad-hoc or exploratory testing) without losing track of the objectives. For a small, agile team, forcing a rigid and lengthy planning phase slows down the actual development process. Directly executing tests and documenting the results is a much more efficient and practical use of our limited time and resources.